PROJECT QUALITY MANAGEMENT PLAN
LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

ASIA PACIFIC COLLEGE
3 HUMABON, MAGALLANES
MAKATI CITY, METRO MANILA 1232

APRIL 1, 2026

PROJECT MANAGEMENT DOCS

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INTRODUCTION

The Quality Management Plan is an integral part of the LitPath AI project management plan. The purpose of this plan is to describe how quality will be managed throughout the software development lifecycle (SDLC), from initial architecture design through deployment, testing, and post-launch maintenance. It includes the processes and procedures for ensuring quality planning, assurance, and control are consistently executed. All stakeholders, developers, and QA personnel should be familiar with how quality will be planned, assured, and controlled.

The Quality Management Plan for the LitPath AI project will establish the activities, processes, and procedures for ensuring a secure, accurate, and high-performing software product upon the conclusion of the project. The purpose of this plan is to:

Ensure quality is planned into every SDLC phase and AI pipeline component.

1. Define how quality will be managed across development, testing, and deployment.
2. Define quality assurance activities focused on process compliance, code reviews, and architecture validation.
3. Define quality control activities focused on functional testing, AI accuracy validation, performance benchmarking, and security verification.
4. Define acceptable quality standards aligned with DOST-STII guidelines, Philippine Data Privacy standards, and software engineering best practices.

QUALITY MANAGEMENT APPROACH

This section describes the approach the organization will use for managing quality throughout the project's life cycle. Quality must always be planned into a software project to prevent rework, technical debt, security vulnerabilities, and inaccurate AI output. Quality will be considered from both a **product** (software functionality, AI relevance, UX, performance) and **process** (SDLC adherence, code quality, documentation, testing coverage) perspective.
The quality management approach for LitPath AI will ensure quality is planned for both the product and processes. To be successful, this project will meet its quality objectives by utilizing an integrated quality approach to define quality standards, measure quality, and continuously improve through Agile retrospectives and AI pipeline evaluations.

Product Quality for LitPath AI will be defined by DOST-STII functional specifications, AI relevance thresholds, security compliance (OWASP, DPA 2012), and user experience standards. The focus is on delivering a stable, accurate, and responsive web application that meets established quality standards and user satisfaction.

Process Quality for LitPath AI will focus on the development of workflows, version control practices, CI/CD pipeline execution, documentation compliance (aligned with the DOST-STII SDS

template), and testing protocols. Establishing process quality standards will ensure all development activities conform to organizational engineering standards.

The project team will work with the QA Tester and Project Lead to define and document all organizational and project-specific quality standards for both product and processes. All quality documentation will become part of the LitPath AI Project Plan and will be transitioned to DOST-STII IT operations upon successful project completion.

Metrics will be established and used to measure quality throughout the project life cycle. The QA Tester will be responsible for working with the development team to define these metrics, conduct measurements, and analyze results. These product and process measurements will be used as key criteria in determining project success and must be reviewed by the Project Sponsor. Metrics will include:

1. **Schedule:** Sprint delivery rate, milestone adherence.
2. **Resources:** Developer/QA allocation efficiency, cloud/API utilization.
3. **Cost:** Budget vs. actual spending (hosting, Gemini API usage, development tools).
4. **Process Performance:** Code review turnaround time, CI/CD pipeline success rate, documentation completeness vs. SDS template.
5. **Product Performance:** API response time (<2s standard, <5s streaming), AI relevance score (cosine similarity threshold <1.5), system uptime (≥99.5%), test coverage (≥80% unit/integration).
6. **Customer Satisfaction:** CSM survey average rating (≥4.0/5.0), feedback resolution rate, failed query rate.

Quality improvements will be identified by any team member. Each recommendation will be reviewed to determine cost vs. benefit, impact on AI accuracy, system performance, or user experience. If implemented, the Project Lead will update project documentation, and the QA Tester will update test suites and QA logs.

QUALITY REQUIREMENTS / STANDARDS

This section describes how the project team and QA group will identify and document quality requirements and standards, and how compliance will be demonstrated.

Product Quality:

Product quality standards will be determined by the development team, QA Tester, and DOST-STII stakeholders. Standards will be based on:

- DOST-STII functional and non-functional requirements.
- AI/ML accuracy benchmarks (reranking validation, citation accuracy, query rewriting precision).

- Security & compliance: OWASP Top 10, Philippine Data Privacy Act (DPA) 2012, secure token authentication, password hashing (PBKDF2).
- Performance standards: Streaming response latency, concurrent user handling, database query optimization.

The product will be deemed compliant once it passes UAT, achieves ≥80% automated test coverage, maintains AI relevance scores within acceptable thresholds, and receives sign-off from DOST-STII stakeholders after a 5-day beta trial period.

Process Quality:
Process quality standards will focus on SDLC adherence, Git branching strategy, peer code reviews, CI/CD automation, and documentation alignment with the DOST-STII SDS template. Unique steps in AI pipeline integration (RAG, vector indexing, LLM prompting) will require specific validation protocols.
Process compliance will be achieved when all pull requests undergo mandatory peer review, CI/CD pipelines pass without critical failures, all API endpoints are documented, and the SDS v1.0 is fully approved by the project sponsor.

QUALITY ASSURANCE

Quality Assurance focuses on the **processes** used in developing, testing, and deploying the LitPath AI product. An iterative QA process will be used throughout the SDLC. This includes measuring process metrics, analyzing development data, and continuously improving workflows through sprint retrospectives and pipeline audits.

The Project Lead and QA Tester will perform assessments at planned intervals to ensure all development processes are correctly implemented. Key process metrics include code review completion rate, CI/CD build success rate, documentation adherence, and AI pipeline validation accuracy.

Process Action	Acceptable Process Standards	Process Phase	Assessment Interval
Code Review & Merge	100% PRs require ≥1 peer approval: zero critical linting/security issues	Development	Per sprint / per merge
CI/CD Pipeline Execution	≥95% automated build/test success rate; rollback protocol documented	Build & deploy	Daily / per commit
AI Pipeline Validation	RAG components (embedding, reranking, generation) pass	AI Integration	Weekly

	benchmark tests with ≥85% relevance accuracy		
Documentation Compliance	SDS, API docs, and architecture diagrams updated and approved per DOST-STII template	Planning & Design	Bi-weekly
Security & Data Privacy Audit	Token auth, password reset, and CSM consent flows pass OWASP & DPA checks	Testing & pre-launch	Monthly / Pre-release

The QA Tester will provide day-to-day quality management, conduct process audits weekly, monitor development metrics, and ensure all processes comply with project and organizational standards. If discrepancies are found, the QA Tester will meet with the Project Lead to review findings and initiate corrective actions.

The Project Lead will schedule regularly sprint reviews, management check-ins, and documentation audits. Agenda items will include process compliance, audit findings, and process improvement initiatives.

Process improvement is a core aspect of QA. All findings should result in documented workflow optimizations, test automation updates, or AI prompt refinements. Changes will be communicated to all stakeholders.

QUALITY CONTROL

Quality Control applies to the **product** (LitPath AI application) and focuses on monitoring, testing, and recording results to assess performance and recommend necessary changes. It verifies that deliverables meet defined standards before release.

QC for LitPath AI will focus on functional accuracy, AI response quality, UI/UX consistency, API performance, and security resilience. All trial builds and staged releases will undergo structured testing by the QA Tester.

Product / Module	Physical/Performance Standards	Quality Assessment Activities	Assessment Intervals
Auth & Session Management	Token generation/expiration works correctly; PBKDF2 hashing verified; zero auth bypass vulnerabilities	Unit testing, security scanning, manual penetration tests	Per sprint / Pre-release
Search & RAG Pipeline	AI answers citation-backed; cosine similarity <1.5; reranking precision ≥80%;	AI evaluation benchmarks, load testing, manual	Weekly / Pre-release

	streaming latency <5s	query validation	
Admin Dashboard & Analytics	KPI calculations accurate; charts render correctly; date filters apply to all tables	E2E testing, data reconciliation with DB, cross-browser testing	Bi-weekly
CSM Feedback & Compliance	Form validates required fields; data privacy consent logged; responses stored securely in Supabase & PostgreSQL	Functional testing, DPA compliance review, UAT	Monthly / Post-deployment
API Endpoints (REST)	All endpoints return correct HTTP status codes; JSON schemas match docs; rate limiting applied	Automated API testing (Postman/PyTest), contract testing	Per deployment

The QA Tester will execute all test suites, log defects, and verify fixes. Developers will resolve critical/high-priority bugs within 48 hours. The QA Tester will ensure all physical and performance standards are met, perform audit trails, and maintain QC documentation.

The Project Lead will schedule product review meetings to discuss test results, defect trends, and product improvement initiatives. Meeting these standards ensures LitPath AI achieves high user satisfaction and aligns with DOST-STII operational expectations.

Quality Control Measurements

All LitPath AI products and processes must be measured and meet established standards. The logs below will be used by the development and QA teams to document findings, track resolutions, and serve as acceptance evidence.

Quality Assurance Log

Sprint/Date	Process Measured	Required Value	Actual Measured	Acceptable? (Y/N)	Recommendation	Date Resolved
	Code Review Compliance	100% PRs approved				
	CI/CD Success Rate	≥95%				
	AI Pipeline Relevance	≥85% accuracy				

	Documentation Update	SDS & API docs aligned				

Quality Control Log

Build/ Date	Component Tested	Required Standard	Actual Result	Acceptable? (Y/N)	Recommendation	Date Resolved
	Auth Token Flow	24h expiry, 1 session/user				
	RAG Search Query	<2s response, relevant sources				
	Streaming Endpoint	<5s latency, no truncation				
	CSM Form Submission	DPA consent logged, DB sync				
	Admin KPI Dashboard	Accurate counts, filters work				

If actual measurements do not meet standards, corrective actions will be documented, prioritized in the backlog, and resolved before the next release candidate. Discrepancies will be reviewed in sprint retrospectives or ad-hoc quality meetings as necessary.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

_____ Date: April 1, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)